

GYAN SETU

Chapter-1, Number System Notes

Numbers

1. Natural Number

Definition: Counting numbers starting from 1.

Examples: 1, 2, 3, 4, 5, ...

2. Whole Numbers

Definition: All natural numbers including 0.

Examples: 0, 1, 2, 3, 4, ...

3. Odd Numbers

Definition: Numbers that are not divisible by 2.

Examples: 1, 3, 5, 7, 9, ...

4. Even Numbers

Definition: Numbers that are divisible by 2.

Examples: 0, 2, 4, 6, 8, ...

5. Prime Numbers

Definition: Numbers that have only two factors: 1 and itself.

Examples: 2, 3, 5, 7, 11, 13, ...

Note: 2 is the only even prime number.

6. Composite Numbers

Definition: Numbers that have more than two factors.

Examples: 4, 6, 8, 9, 10, 12, ...

Note: 1 is neither prime nor composite.

7. Integers

Definition: All positive numbers, negative numbers, and 0 (but no fractions or decimals).

Examples: -3, -2, -1, 0, 1, 2, 3, ...

8. Rational Numbers

Definition: Numbers that can be written in the form p/q , where $q \neq 0$ and p, q are integers.

Examples: $1/2$, $-3/4$, 5, 0.25 ($=1/4$)

9. Irrational Numbers

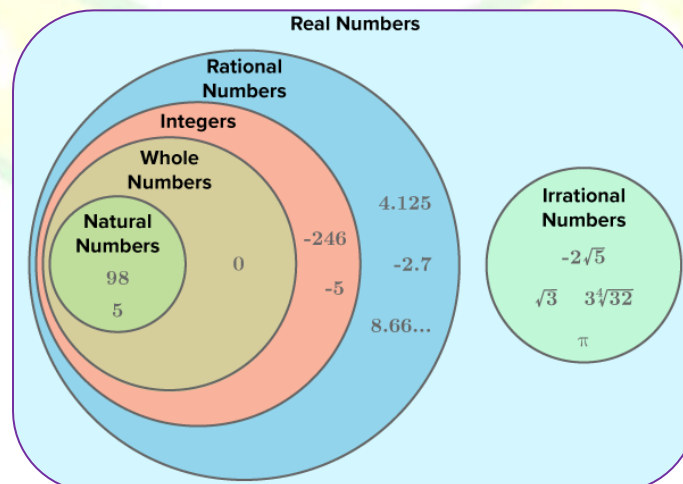
Definition: Numbers that cannot be written as a fraction p/q . Their decimal form is non-repeating and non-terminating.

Examples: $\sqrt{2}$, π , $\sqrt{3}$

10. Real Numbers

Definition: All rational and irrational numbers together are called real numbers.

Examples: -5, 0, $3/4$, $\sqrt{2}$, π



Important Note:

If r is rational and s is irrational, then $r + s$ and $r - s$ are irrational numbers, $(r.s)$ and (r/s) are irrational numbers, $r \neq 0$.

For e.g.:- Let $r = 2$ and $s = \sqrt{2}$

Where r is rational and s is irrational

Therefore, $r + s = 2 + \sqrt{2} \Rightarrow$ Irrational

$r - s = 2 - \sqrt{2} \Rightarrow$ Irrational

$r . s = 2 . \sqrt{2} \Rightarrow$ Irrational

$r / s = 2 / \sqrt{2} \Rightarrow$ Irrational

Important Identities of Real Number:

- 1) $\sqrt{(a . b)} = \sqrt{a} . \sqrt{b}$
- 2) $\sqrt{(a / b)} = \sqrt{a} / \sqrt{b}$
- 3) $\sqrt{[(a + b) . (a - b)]} = a - b$
- 4) $(a + \sqrt{b}) . (a - \sqrt{b}) = a^2 - b$
- 5) $(\sqrt{a} + \sqrt{b})^2 = a + 2 . \sqrt{a} . \sqrt{b} + b$

Laws of Exponents:

Let $a > 0$ be a real number and p and q be rational numbers. Then:

- 1) $a^p . a^q = a^{p+q}$
- 2) $a^p / a^q = a^{p-q}$
- 3) $(a^p)^q = a^{p.q}$
- 4) $a^p . b^p = (a . b)^p$